

CHAPTER: UNIQUE KEY

What is a Unique Key?

A Unique Key is a constraint that ensures all values in a column (or combination of columns) are unique.

No two rows can have the same value for a Unique Key.

A Unique Key:

- Prevents duplicate values.
- Can consist of one or more columns.
- Can have multiple constraints in one table.
- Is commonly used to enforce business rules.

NULL Handling

This is one of the most frequently asked interview topics.

Can a Unique Key contain NULL?

Answer: Yes, but the exact behavior depends on the DBMS.

Why?

NULL means Unknown value. Unknown values are not considered equal under the SQL standard.

Because of this, many databases allow multiple NULL values in a UNIQUE column.

Example:

Email
aadya@gmail.com
NULL
NULL

This is accepted by many database systems.

DBMS Support for Multiple NULLs

Database	Multiple NULLs in UNIQUE?
MySQL	✓ Yes
PostgreSQL	✓ Yes
Oracle	✓ Yes
SQL Server	⚠ Traditionally only one NULL in a UNIQUE constraint (behavior depends on implementation and indexing options)

Multiple Unique Keys in a Table

a table can have many Unique Keys.

Possible constraints:

- PRIMARY KEY(EmployeeID)
- UNIQUE(Email)
- UNIQUE(Phone)
- UNIQUE(Username)

SQL Examples

Creating a Unique Key

```
CREATE TABLE Employee
(
    EmployeeID INT PRIMARY KEY,

    Email VARCHAR(100) UNIQUE,

    Name VARCHAR(50)
);
```

Multiple Unique Keys

```
CREATE TABLE UserAccount
(
    UserID INT PRIMARY KEY,

    Email VARCHAR(100) UNIQUE,

    Username VARCHAR(50) UNIQUE,

    Phone VARCHAR(20) UNIQUE
);
```

Composite Unique Key

```
CREATE TABLE Student
(
    RollNo INT,

    Semester INT,

    UNIQUE(RollNo, Semester)
);
```

This prevents duplicate (RollNo, Semester) combinations.

Relationship to Foreign Key

A Foreign Key may reference a Unique Key (not just a Primary Key), because it also guarantees uniqueness.

Disadvantages

Disadvantage 2 — Business Rules May Change

Suppose Phone Number is unique today.

Tomorrow, the company allows family members to share one phone number.

The Unique constraint must be removed.

Disadvantage 3 — Additional Indexes

Most DBMSs create an index to enforce a Unique constraint.

Having many Unique Keys can increase storage and slow down insert/update operations because additional indexes must be maintained.